

DentalVision: Automated Tooth Segmentation for Dental Diagnostics using Vision Transformers

Submitted By

Abhishek Kumar Singh (M23CSA503)
m23csa503@iitj.ac.in



M.Tech. - Artificial Intelligence

Indian Institute of Technology Jodhpur
Department of Computer Science and Engineering

Abstract

Dental X-ray images contain valuable information about tooth structures, but identifying and describing these structures manually can be time-consuming and dependent on expert annotation. This project presents DentalVision, a deep-learning prototype for automated tooth-region segmentation from dental X-ray images. The system uses the Humans in the Loop Teeth Segmentation on Dental X-ray Images dataset, consisting of X-ray images and corresponding human-annotated masks.

A custom ViT-CNN architecture is developed using a Vision Transformer encoder and a convolutional upsampling decoder. Each input image is converted to RGB, resized to 224 x 224 pixels, and represented as non-overlapping 16 x 16 patches. The transformer produces 196 spatial patch tokens and one class token. The class token is removed, and the patch tokens are reshaped into a 14 x 14 feature map before being processed by the decoder. Convolutional and transpose-convolutional layers reconstruct a one-channel segmentation map, which is converted into a binary mask using a sigmoid activation and a threshold of 0.5. The training workflow uses binary cross entropy loss, the Adam optimizer, a learning rate of 1e-4, and a batch size of 8. The committed training script runs for 5 epochs over the full set of annotated images without a held-out validation split, and the model state saved at the end of the final epoch is the checkpoint used for inference.

For practical demonstration, the trained model is integrated into a Streamlit application that accepts JPG and PNG images and displays the predicted mask. The project demonstrates an end-to-end workflow from dataset preparation and model training to interactive inference. It should be considered a research and educational prototype rather than a clinically validated diagnostic system. Because the current training script holds out no data, the quantitative segmentation quality of the model has not been established, and establishing it is the first item of future work. Further work includes a reproducible train-validation-test split with per-epoch Dice and IoU reporting, best-checkpoint selection, comparison with standard U-Net baselines, improved spatial reconstruction, uncertainty visualization, and investigation of hybrid quantum-classical or QUBO-based segmentation refinement.

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1 Objective

The objective of this project is to develop an automated tool that segments teeth in dental X-ray images using deep learning techniques. The tool should provide accurate visualizations and be accessible through a user-friendly web application to assist dentists in diagnostics and treatment planning.

2 Approaches Used

- **Model:** Vision Transformer (ViT) as the encoder and a custom lightweight U-Net-inspired decoder.
- **Data Preprocessing:** Images and masks resized to 224x224; grayscale normalization; binary thresholding.
- **Training:** Binary Cross Entropy loss with Adam optimizer, trained using PyTorch and Hugging Face's `transformers` library.
- **Interface:** Developed an interactive web app using Streamlit.

3 Dataset

- **Source:** Kaggle - Teeth Segmentation Dataset.
- **Content:** Panoramic dental X-rays and binary masks labeled by humans.
- **Format:** Separate folders for images and masks, available in PNG and JSON format.

4 Model Architecture

We use a Vision Transformer (ViT) as an encoder and a custom decoder inspired by U-Net to perform segmentation.

- **Encoder:** ViT with patch size 16, hidden size 768, and 6 transformer layers.
- **Decoder:** A stack of Conv2D, ReLU, and ConvTranspose2D layers for upsampling.
- **Output:** 1-channel sigmoid-activated mask for binary segmentation.

5 Training and Evaluation

The following settings are taken directly from the committed training notebook, `src/DentalVision_CV_Pro`.

- **Loss Function:** Binary Cross Entropy (`nn.BCELoss`, applied to sigmoid outputs).
- **Optimizer:** Adam (learning rate = 0.0001).
- **Batch Size:** 8.
- **Epochs:** 5.
- **Train-Validation Split:** None. A single `DataLoader` iterates over the full dataset, so no data is held out.
- **Checkpoint Selection:** None. The model state after the final epoch is saved and deployed.
- **Training Loss:** 0.5152 at epoch 1, decreasing to 0.3592 at epoch 5.

Because no data is held out, the training script computes no validation or test metrics. Segmentation quality is therefore reported qualitatively in Section 6, and quantitative evaluation is deferred to future work.

6 Results

Training loss decreased monotonically across all five epochs, from 0.5152 to 0.3592, indicating that the model learned a non-trivial mapping rather than collapsing to a constant prediction. For reference, a model predicting a uniform probability of 0.5 at every pixel would yield a binary cross entropy of approximately 0.693.

Qualitatively, the trained model produces coherent tooth-region masks on the sample images included in the repository, and the deployed application returns a plausible mask for uploaded panoramic X-rays.

The following limitations apply to these results and should be read alongside them:

- No validation or test set was held out, so no Dice, IoU, or other overlap metric has been measured for this model.
- Because every available image was used for training, no claim can be made about generalization to unseen X-rays.
- Training loss alone is not a sufficient indicator of segmentation quality, particularly under foreground-background class imbalance.

7 Deployment Link

The model is deployed and accessible through a live Streamlit app:

Live App: <https://dentalvision-vitonet.streamlit.app>

8 GitHub Repository

The complete source code and training scripts are available on GitHub:

Repository URL: <https://github.com/akshziitj/dentalvision-vitonet>

9 Screenshots from Deployed Website

The deployed application provides an upload interface for dental X-ray images. An example extracted from the project report is shown below.



Figure 1: Upload interface for dental X-rays.

The project report also presents the results as the original image, mask prediction, and overlay.



Figure 2: Results: Original image, mask prediction, and overlay.

10 Model Saving and Inference

The trained model is saved as `vit_teeth_segmentation.pth` and can be loaded for predictions.

```
model.load_state_dict(torch.load('vit_teeth_segmentation.pth'))
```

11 Project File Structure

The project structure is organized as follows:

```
dentalvision-vitonet/
|-- src/                                # Source code and training pipeline
|   |-- DentalVision_CV_Project.ipynb # Training notebook
|   |-- streamlit_app.py              # Streamlit web application
|   |-- utils.py                      # Model definition and inference helpers
|   |-- requirements.txt              # Python dependencies
|-- datasets/                          # Teeth Segmentation dataset (images, masks
    , JSON)
|-- docs/                              # Report and presentation
|-- DentalVision_LaTeX_Project/        # LaTeX sources for this report
|-- sample_data/                       # Sample input and predicted mask
|-- .gitattributes                     # Git LFS configuration for *.pth
```

-- .gitignore	# Files and folders excluded from git
-- README.md	# Project overview and instructions

The trained checkpoint `vit_teeth_segmentation.pth` is tracked with Git LFS and is not committed to the repository in its current state; it must be supplied separately before the application can run.

12 Conclusion

DentalVision demonstrates how a Vision Transformer encoder can be combined with a convolutional decoder to perform binary tooth-region segmentation, and how such a model can be delivered through an interactive web interface. The project establishes a complete pipeline from dataset preparation through training to deployment. It remains a research and educational prototype: without a held-out evaluation set its segmentation accuracy is unquantified, and it is not suitable for clinical use.

13 Future Work

- Introduce a reproducible train-validation-test split and report Dice and IoU on a held-out test set. This is a prerequisite for any performance claim.
- Add per-epoch validation and select the deployed checkpoint by best validation Dice rather than by final epoch.
- Initialize the encoder from pretrained ViT weights instead of random initialization, with the corresponding input normalization.
- Introduce data augmentation appropriate to panoramic radiographs.
- Compare against a standard U-Net baseline on an identical split.
- Integrate instance segmentation to identify individual teeth, which the source annotations already support.
- Enable detection of anomalies (cavities, missing teeth, etc.).
- Enhance the model with Swin Transformers or SAM.

14 Acknowledgments

- **Kaggle:** For the open-source dental X-ray dataset.
- **Hugging Face:** For the Vision Transformer model base.
- **Streamlit:** For the lightweight UI framework.

15 References

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